

Ziyi Yang (杨子毅)

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Expected Graduation: 2028-06

Date of Birth.: 2001-04



Education

Ph.D.	Organic Chemistry	Tsinghua University	2023 – Now
B.S.	Chemical Biology	Tsinghua University	2019 – 2023

Research: Protein Design & Synthesis

I aim to explore the **beyond biology proteins** with the power of AI and Chemistry.

1) Cyclic Peptide Binder Design

I built million-scale synthetic dataset on cyclic peptide – protein complex structures by mimicking inter-chain interfaces from intra-protein interactions and atom-level CFG, enabling model scaling and user-define generation.

Z. Yang, H. Xie, Y. Jia, et al. *NeurIPS*, **2025**.

Z. Yang[†], Y. Jia[†], G. Deng[†], et al. *NeurIPS*, **2026**.

2) Mirror-image Peptide Binder Design

I introduced axial-vector / pseudo-scalar features to explicitly encode chirality, enabling chirality control and generalization for hetero-chiral protein designs.

Z. Yang[†], Z. Tian[†], Y. Jia[†], et al. *ICML*, **2026**.

3) Glycosylation for Protein Folding

I participated a project that shows glycosylation could prevent aggregation during folding, enabling the synthesis of hard-to-fold proteins.

T. Wang[†], W. Shi[†], **Z. Yang**, et al. *Angew. Chem.*, **2024**.

4) Synthesis of Ultra-Large Proteins

I participated a project that invented a new ligase that is able to ligate two large protein segments, enabling the total synthesis of a > 100kDa protein.

Manuscript in preparation

Visit and Intern

2026.07 – Now	LigIT AI	Intern
2025.03 – Now	Yanyan Lan Lab (AIR, Tsinghua)	Intern
2026.04 – 2026.06	Bytedance AI Drug Discovery	Intern
2023.06 – 2023.07	Jishen Zheng Lab (USTC)	Summer Visit
2022.07 – 2023.08	Peilong Lu Lab (Westlake Univ.)	Summer Visit